Superpolynomial online mistakes for isotropic Bayesian hardtanh networks

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

We construct small Boolean targets and fixed nonrepeating input sequences on which exact Bayesian learning with a one-hidden-layer hardtanh network makes superpolynomially many online mistakes. The weight prior is independent Gaussian and isotropic within each layer. It is conditioned from the start on an absolute output margin at every Boolean input, and training conditions on exact labels. At every sufficiently large fixed learner width, the correct-label posterior probability tends to zero on each designated round as the Gaussian scale tends to zero.

The targets have same-depth hardtanh teachers of width C, margin one, and coefficients bounded by C, while the construction forces $\exp(\Omega(C^{1/3}))$ designated mistakes. The polynomial exponent is a convenient bound, not an optimization claim. The calibration prefix and sufficient learner width are large.

The proof gives an exact unequal-rail parity norm for hardtanh neurons and polynomial stability into an odd Fourier subspace. A shared carrier bit converts hardtanh coefficients into even-degree ReLU coefficients. Cauchy–Schwarz and a monotone finite-difference estimate then control all affine neurons. The resulting parity costs have positive inverse-degree curvature, so the cheapest competing scores use a fixed negative overshoot. An explicit dual identifies these scores, square-root independence orders their costs, and integer neuron replication transfers the strict cost gaps to the actual Gaussian posterior.

1 Model and result

Write $\mathrm{clip}(t) = \max(-1, \min(1, t))$. On the full Boolean cube $X = \{-1, 1\}^d$, append a constant input $\bar{x} = (1, x)$ and use

$$F_\theta(x) = \sum_{j=1}^{N} v_j \,\mathrm{clip}(u_j \cdot \bar{x}), \qquad (u_j)_i \sim \mathcal{N}(0, N\sigma^2/d), \qquad v_j \sim \mathcal{N}(0, \sigma^2/N),$$

with all coordinates independent. The prior is first conditioned on $|F_\theta(x)| > 1/2$ at every input. Each revealed example then conditions on its exact sign label. The strong-prior limit means $\sigma \downarrow 0$ after fixing N; no joint width–scale limit is used. Up to the common factor $1/(2\sigma^2)$, the Gaussian energy is

$$K_N(\theta) = \frac{d}{N} \sum_j \|u_j\|_2^2 + N \sum_j v_j^2. \tag{1}$$

*Authors are listed in alphabetical order.

Theorem 1. *For all sufficiently large integers m, there exist a dimension $d_m = \Theta(m^2)$, a Boolean function f_m on the full cube, and a fixed nonrepeating input sequence with $T_m = \exp(\Omega(m))$ designated rounds, with the following properties. The target has a one-hidden-layer hardtanh score of margin one and width $C_m = O(m^3)$, all of whose coefficients are bounded by C_m. There is a finite $N_0(m)$ such that, for every fixed $N \geq N_0(m)$, at every designated round the posterior probability of the correct label tends to zero as $\sigma \downarrow 0$. Consequently the same sequence forces $\exp(\Omega(C_m^{1/3}))$ mistakes for posterior-majority prediction in this limit, and that many expected mistakes under posterior sampling.*

This is an adversarial-order result. It does not contradict iid sample-complexity guarantees for small targets. The full-cube margin restriction and the order of limits are part of the theorem, as are the large calibration prefix and the potentially very large width threshold. The proof concerns the full Bayesian posterior, rather than a chosen minimum-energy representative.

2 The parity code and its scalar costs

Take ℓ pairs of disjoint rails B_j^0, B_j^1 of even sizes, nonzero columns $a_j \in \mathbb{F}_2^m$ spanning the logical space, and one additional physical carrier coordinate $x_\star$. For $s \in \mathbb{F}_2^m$ put

$$A_s = \{\star\} \cup \bigcup_{j=1}^{\ell} B_j^{a_j \cdot s}, \qquad n_s = |A_s|.$$

All n_s are odd. Let $b = x_\star \prod_j \chi_{B_j^0}$, define $(-1)^{\zeta_j} = \chi_{B_j^0 \cup B_j^1}$, and put $z = \sum_j a_j \zeta_j$. Then z is uniform on $G = \mathbb{F}_2^m$ and $\chi_{A_s} = b(-1)^{s \cdot z}$. The full independent-rail code is the special case $\ell = m$, $a_j = e_j$; the construction below retains a path subcode of a larger rail family. The target is

$$f(x) = b(x) \begin{cases} +1, & z(x) = 0, \\ -1, & z(x) \neq 0. \end{cases} \tag{2}$$

For odd n define

$$\kappa_n = 2^{1-n} \binom{n-1}{(n-1)/2}, \qquad w(n) = \frac{\sqrt{n}}{\kappa_n}. \tag{3}$$

A hardtanh neuron with unit weights on n coordinates is majority on those coordinates. Its top parity coefficient has absolute value κ_n, so signed averaging over input flips represents parity with variation cost $w(n)$. For $F = \sum_j c_j \operatorname{clip}(a_j \cdot \bar{x})$, the representation cost is $\sum_j |c_j| \|a_j\|_2$; its infimum is the hardtanh variation norm $V(F)$. Arithmetic–geometric mean gives

$$K_N \geq 2\sqrt{d} \sum_j |v_j| \|u_j\|_2 \geq 2\sqrt{d}\, V(F_\theta). \tag{4}$$

For a logical score $g(z) = \sum_s \widehat{g}(s)(-1)^{s \cdot z}$, write

$$B_w(g) = \sum_s w(n_s) |\widehat{g}(s)|, \qquad W(v) = 2^{-m} \sum_s (-1)^{s \cdot v} w(n_s).$$

The atomic theorem in the next section shows that $V(bg) = B_w(g)$. The following scalar facts identify and order the competing scores.

2.1 Positive curvature and a dominant pair

The gamma-ratio identity and its right-half-plane expansion give

$$w(z) = \sqrt{\pi z}\,\frac{\Gamma((z+1)/2)}{\Gamma(z/2)} = \sqrt{\frac{\pi}{2}}\left(z - \frac{1}{4} + \frac{1}{32z} + O(|z|^{-2})\right). \tag{5}$$

The remainder is holomorphic and uniform after restricting to a far right half-plane. If $C = \sqrt{\pi/2}/32$, Cauchy's estimate on the circle of radius $a(1-1/k)$ gives

$$\frac{Ck!}{2a^{k+1}} \le (-1)^k w^{(k)}(a) \le \frac{3Ck!}{2a^{k+1}}$$

for $2 \le k \le \ell$ and $a \ge n_0 \gg \ell^2$. On the full independent-index cube, if $n_s = n_0 + \sum_i \delta_i s_i$ and $D_\delta = \sum_i \delta_i$, then

$$\frac{Ck!\prod_{i\in T}\delta_i}{2^{k+1}(n_0+D_\delta)^{k+1}} \le W(T) \le \frac{3Ck!\prod_{i\in T}\delta_i}{2^{k+1}n_0^{k+1}} \qquad (|T| = k \ge 2). \tag{6}$$

The degree-one coefficients are negative, since $w' > 0$.

The same estimates apply before pushing the independent rail indices through the column map. In particular, higher-order coefficients are positive on the independent-index cube; their sums give coefficients of the retained column code. The only possibly negative contributions come from individual columns. Section 4 uses this observation to build and order the negative-overshoot competitors.

3 An exact unequal-rail hardtanh norm

In this section consider the full family of transversals, indexed by $s \in \mathbb{F}_2^\ell$ independently, before retaining the column subcode. Write $m = \ell$ temporarily. Let $\mathcal{C}$ be the span of the code characters χ_{A_s}. Let $\mathcal{D}_{\mathrm{odd}}$ be the span of all odd-degree characters whose supports are contained in some A_s. The odd-degree restriction is essential: a complete rail transversal without the carrier has even degree and would otherwise defeat the observability argument used later.

Theorem 2 (Sharp atom bound and support stability)**.** *Suppose $m \ge 2$, all rail sizes are even and lie in $[R, R+\Delta]$, $\Delta \ge 2$ is even, and $R \ge 6\Delta$. Every affine hardtanh atom $\phi_u(x) = \mathrm{clip}(u \cdot \bar{x})$ satisfies*

$$Q(\phi_u) := \sum_s w(n_s)|\widehat{\phi}_u(A_s)| \le \|u\|_2.$$

Nonzero equality requires zero bias, zero irrelevant and unused-rail weights, and magnitude one on the carrier and every coordinate of a single selected transversal. Moreover

$$\|\phi_u - P_{\mathcal{D}_{\mathrm{odd}}}\phi_u\|_2^2 \le 2^8 R^2 \|u\|_2(\|u\|_2 - Q(\phi_u)). \tag{7}$$

Consequently $V(bg) = B_w(g)$ for every code score.

3.1 Positive Fourier integrals and block balancing

Allow an auxiliary threshold $v > 0$ and write $\mathrm{clip}_v(t) = \mathrm{ReLU}(t+v) - \mathrm{ReLU}(t-v) - v$. A code coefficient of clip_v is twice the ReLU coefficient obtained by adjoining a Boolean sign ξ of weight v

and including it in the support. Every augmented support has even degree. The even-degree ReLU Fourier integral, up to its fixed degree sign, is

$$\frac{1}{\pi}\int_0^\infty \cos(bt)\prod_{j\in A}\sin(u_j t)\prod_{j\notin A}\cos(u_j t)\,\frac{dt}{t^2}. \tag{8}$$

Discard the absolute bias and irrelevant cosine factors, flip signs of incoming weights, and take absolute values inside this integral. Call the resulting weighted expression $H(u_0, \mathrm{rails}; v)$, where $u_0 \geq 0$ is the physical carrier magnitude. It bounds Q and is homogeneous of degree one jointly in its arguments. The auxiliary weight is excluded from the physical incoming norm.

Hölder on each even rail, applied to the disjoint union of coefficient-index copies of the integration measure, bounds H by the geometric mean of its values on vectors obtained by choosing one coordinate magnitude uniformly in each rail and repeating it throughout that rail. The carrier and auxiliary weights stay fixed. This statement concerns the positive absolute-integral expression, not an equality between signed coefficients. Sampling preserves the expected squared physical norm. Thus a norm bound for these balanced potentials transfers to arbitrary coordinates by Jensen's inequality.

For balanced rails name their amplitudes $a_i \geq b_i \geq 0$, with respective sizes p_i, q_i, and put $D = \sum_i p_i$, $n = D + 1$. Let $J(c)$ be the weighted ReLU coefficient sum with two unpaired shared coordinates, both of magnitude c, and the actual rail weights. Every sine and cosine exponent in its integral is even, so the integral has no sign cancellation. Cauchy–Schwarz on the two shared factors in H gives

$$H(u_0, \mathrm{rails}; v) \leq 2\sqrt{J(u_0)J(v)}. \tag{9}$$

3.2 A randomized minimum for unequal rails

Write $\kappa_D^{\mathrm{R}} = 2^{1-D}\binom{D-2}{(D-2)/2}$ for even D. Consider a zero-bias ReLU ridge with selected rail coordinates X_i, opposite coordinates Y_i, and any additional unpaired even block. Let c_s be its absolute coefficient on the transversal choosing Y_i when $s_i = 1$, together with the unpaired block. Put $T_\varepsilon = \min_i(a_i + \varepsilon_i b_i)$.

Lemma 3 (Unpaired shared block)**.** *If the unpaired block consists of two coordinates of weight c, then*

$$\sum_s c_s \leq \kappa_{D+2}^{\mathrm{R}}\,\mathbb{E}\min(c, T_\varepsilon).$$

If $q_i \geq p_i$, the coefficient mass using the opposite rail in pair i is at most $\kappa_{D-p_i+q_i+2}^{\mathrm{R}} b_i \leq \kappa_{D+2}^{\mathrm{R}} b_i$.

Proof. The fixed signs of the even-block integrals express the unweighted sum as a top selected-parity coefficient after conditioning

$$\chi_{Y_i} = \eta_i \chi_{X_i}, \qquad \eta_i = (-1)^{(q_i - p_i)/2},$$

for every pair. To generate each conditional law, choose independent uniform signs ε_i, set a distinguished opposite coordinate $Y_{i,1} = \varepsilon_i X_{i,1}$, and condition the remaining opposite coordinates on $\chi_{Y_{i,\mathrm{rest}}} = \eta_i \varepsilon_i \chi_{X_{i,\mathrm{rest}}}$. The distinguished selected coordinate now has nonnegative coefficient $a_i + \varepsilon_i b_i$. The conditional laws do not depend on any of these distinguished selected coordinates.

The normalized finite derivative of ReLU in a distinguished coordinate is a nondecreasing function into $[0, 1]$. It remains nondecreasing in all other selected coordinates under these conditional laws: changing an ordinary selected bit increases its direct contribution by $2a_j$ and reverses one opposite

parity condition. Couple the two uniform parity classes by flipping one fixed remaining opposite bit, changing the total argument by $2a_j - 2b_j Y_{j,k} \geq 0$. An unpaired shared bit simply increases the argument by $2c$.

A nondecreasing function on k Boolean coordinates with values in $[0,1]$ has top Fourier coefficient at most $2^{-k}\binom{k-1}{\lfloor (k-1)/2 \rfloor}$ in absolute value. Indeed average over permutations and express the resulting increasing Hamming-level sequence as a convex combination of level thresholds; their alternating binomial sums give this bound. Apply it after taking a finite derivative in any distinguished selected bit or in either shared bit. At fixed ε the top coefficient is at most $\kappa^{\mathrm{R}}_{D+2} \min(c, T_\varepsilon)$. Averaging proves the first assertion.

For the marginal assertion, discard the cosine factor of the unused selected rail X_i in the positive integral. Treat Y_i as an unpaired selected block, and take its distinguished derivative of coefficient b_i. The same argument gives $\kappa^{\mathrm{R}}_{D-p_i+q_i+2} b_i$. Finally $\kappa^{\mathrm{R}}_{r+2}/\kappa^{\mathrm{R}}_r = (r-1)/r < 1$. □

3.3 The exact norm bound

The hardtanh cost $w(n)$ is increasing and its odd forward slopes are at most three. Indeed

$$\frac{w(n+2)}{w(n)} = (1+1/n)\sqrt{1+2/n} \leq (1+1/n)^2.$$

The elementary central-binomial bound $w(n) \leq 2n$ therefore gives $w(n+2) - w(n) \leq 4 + 2/n \leq 6$. For completeness, the bound follows by induction from $\kappa_{2r+1} \geq 1/(2\sqrt{r})$ for $r \geq 1$, using the recurrence $\kappa_{2r+3}/\kappa_{2r+1} = (2r+1)/(2r+2)$; $n = 1$ is immediate.

Set $t = \mathbb{E} T_\varepsilon$ and $E_a = \sum_i (a_i - t)$. Monotonicity and the slope bound imply, even when some opposite rails are shorter,

$$w(n_s) \leq w(n) + 3 \sum_{i : q_i > p_i} (q_i - p_i) s_i.$$

Lemma 3 consequently yields

$$J(c) \leq \frac{\sqrt{n}}{2} \mathbb{E} \min(c, T_\varepsilon) + 3\kappa^{\mathrm{R}}_{D+2} \Delta \sum_i b_i.$$

Cauchy–Schwarz, arithmetic–geometric mean, and concavity of the minimum give

$$H \leq J(u_0) + J(v) \leq \frac{\sqrt{n}}{2} [\min(u_0, t) + t] + 6\kappa^{\mathrm{R}}_{D+2} \Delta \sum_i b_i.$$

For every $u_0, t \geq 0$,

$$\frac{n}{2} [\min(u_0, t) + t] \leq Dt + u_0.$$

If $u_0 \geq t$ this is immediate; otherwise the difference equals $(D-1)(t-u_0)/2$. Also, for $i \neq j$,

$$t \leq \frac{a_i + a_j - \max(|a_i - a_j|, b_i)}{2},$$

so summing over ordered pairs gives $\sum_i b_i \leq 2E_a$. The recurrence shows $\kappa_n \sqrt{n} \leq 1$ for odd n; thus $\kappa^{\mathrm{R}}_{D+2}\sqrt{n} \leq 1/2$. Combining these facts and $R \geq 6\Delta$ gives the stronger linear bound

$$H \leq \frac{Dt + u_0 + 6\Delta E_a}{\sqrt{n}} \leq L := \frac{\sum_i p_i a_i + u_0}{\sqrt{n}} \leq \|u_{\mathrm{physical}}\|_2. \tag{10}$$

The quantity L is the projection onto the unit balanced ray on the selected rails and physical carrier. If u_* is that projection,

$$\|u - u_*\|_2^2 = \|u\|_2^2 - L^2 \le 2\|u\|_2(\|u\|_2 - H). \tag{11}$$

Block balancing and Jensen now prove the global atom inequality. Equality forces zero unused weights and a common magnitude a on all selected physical coordinates and the carrier. Equality in the shared-factor Cauchy inequality requires $\sin^2(at)$ proportional to $\sin^2(vt)$ on an interval, hence $a = v$. At the standard threshold $v = 1$ the selected magnitudes are one. Bias and irrelevant weights vanish because they only increase the full incoming norm.

3.4 Polynomial stability into the odd downward space

It suffices to control discarded support weights, rather than to force all retained coordinates to be equal. By joint homogeneity, rescale the auxiliary threshold along with the physical parameters to normalize the original relevant physical norm to one. Let V be its balanced rail sample. If independent rail samples X, Y have second moments $A \ge B$, then

$$\mathbb{E}\min(X^2, Y^2) \ge \frac{AB}{r_{\max}(A+B)} \ge \frac{B}{2r_{\max}}.$$

Choose deterministically the rail with larger second moment in each pair. Its discarded physical squared norm is at most $8R\mathbb{E}d_{\text{face}}(V)^2$, since $R \le r_i^j \le 2R$. For $\epsilon = 1 - H(u) < 1/2$, Hölder and $\mathbb{E}\|V\|^2 = 1$ imply

$$\mathbb{E}\left[1 - \frac{H(V)}{\|V\|}\right] \le -\log H(u) \le 2\epsilon, \qquad \|V\|^2 \le r_{\max} \le 2R.$$

Using (11) bounds the discarded squared norm by $64R^2\epsilon$. For $\epsilon \ge 1/2$, use the trivial bound. Bias and irrelevant squared norm are at most $2\|u\|(\|u\| - Q(\phi_u))$, by the discarded-norm comparison. Strip them and the chosen unused rails. The resulting zero-bias hardtanh atom is odd and depends on one selected support, so lies in $\mathcal{D}_{\text{odd}}$. Hardtanh is one-Lipschitz and the augmented Boolean coordinates have identity covariance. Parameter squared distance therefore bounds the output L^2 residual, proving (7) with the stated safe constant. The exact variation norm follows from the atom inequality and the finite parity-orbit representations.

3.5 Restricting to a linear subcode

The atomic inequality remains valid when only some transversals are retained. The support-stability statement needs a little more care, since its selected full-code transversal might have been omitted. Let $\mathcal{A}$ be any nonempty collection of retained transversals, let $Q_{\mathcal{A}}$ denote the corresponding weighted coefficient sum, and let $\mathcal{D}_{\mathcal{A},\text{odd}}$ be their odd downward Fourier space. For ℓ rail pairs, one has the safe estimate

$$\|\phi_u - P_{\mathcal{D}_{\mathcal{A},\text{odd}}}\phi_u\|_2^2 \le 2^{20}\ell^2R^2\|u\|_2(\|u\|_2 - Q_{\mathcal{A}}(\phi_u)). \tag{12}$$

Here is a proof which does not assume that a balanced hardtanh ray has the optimal gain. Fix one rail B. In the positive absolute integral, sum all full-code terms which use B. For balanced rails, apply the shared-factor Cauchy inequality and the unpaired-block argument of Lemma 3, treating B itself as a fixed selected unpaired block. Discard the cosine factor of its opposite rail. Taking the finite derivative in a coordinate of B bounds each of the two unweighted shared-factor integrals by $\kappa^{\mathrm{R}}_{D'}a_B$, where a_B is its common amplitude and $D' \ge \ell R + 2$. This argument does not

require that B have the larger amplitude in its pair. Since $w_{\max} \le 2(2\ell R+1)$ and $\kappa^{\mathrm{R}}_{D'} \le 1/\sqrt{D'}$, its weighted hardtanh marginal is at most $8\sqrt{\ell R}\, a_B$ (the finitely small endpoint cases are also covered by $R \ge 12$). Block Hölder applied to this marginal, followed by the geometric mean–quadratic mean inequality in the sampled coordinate of B, therefore bounds the arbitrary-coordinate marginal by

$$8\sqrt{\ell R}\left(\frac{1}{|B|}\sum_{j\in B} u_j^2\right)^{1/2} \le 8\sqrt{\ell}\,\|u_B\|_2.$$

Consequently the positive potential over terms using at least one rail from a specified collection of one rail per pair is at most 8ℓ times the Euclidean norm of all those rail weights.

Normalize $\|u\|_2 = 1$ and write $\epsilon = 1 - Q_{\mathcal{A}}(\phi_u)$. The full-code support argument in the preceding subsection applies because the retained coefficient sum is bounded by the full positive potential. It selects one full transversal, and, after removing bias and irrelevant coordinates, its discarded parameter norm ρ satisfies $\rho^2 \le 2^8 R^2 \epsilon$. If that transversal is retained, the one-Lipschitz argument already proves (12). Otherwise every retained transversal uses at least one of these unused rails. The marginal estimate gives

$$1 - \epsilon = Q_{\mathcal{A}}(\phi_u) \le 8\ell\rho \le 2^7 \ell R\sqrt{\epsilon}.$$

For $\epsilon < 1/2$ this implies $\epsilon \ge 1/(2^{16}\ell^2 R^2)$; for $\epsilon \ge 1/2$ the same conclusion is immediate. Approximate by the zero function, which belongs to every retained downward space, and use $\mathbb{E}\phi_u^2 \le \mathbb{E}(u\cdot\bar{x})^2 = 1$. This proves the normalized bound. Rescaling the parameter direction, with the auxiliary hardtanh threshold rescaled inversely, gives the stated homogeneous parameter bound; equivalently repeat the same argument after dividing all positive potentials and parameter norms by $\|u\|_2$.

4 A path code with exponentially many ordered costs

Use r path coordinates together with two distinguished logical coordinates. Thus the logical group is $G = \mathbb{F}_2^{r+2}$. Let a_j be the following $\ell = 2r+1$ columns in this group: the two distinguished unit vectors, all r path unit vectors, and the $r-1$ sums of consecutive path unit vectors. Their respective rail-size increments are

$$\delta_1 = 2Ar, \quad \delta_2 = 4Ar, \quad \delta_{\mathrm{loop}} = 4, \quad \delta_{\mathrm{edge}} = 2,$$

where $A = 2^{20}$ is fixed. Choose a sufficiently large fixed even integer K, and set

$$R = KA^2 r, \quad n_0 = \ell R + 1, \quad S = 6r - 2, \quad D_\delta = 6Ar + S.$$

Each short rail has size R and each long rail size $R + \delta_j$. The retained transversal indexed by $s \in G$ chooses its long rail in pair j exactly when $a_j \cdot s = 1$. Including the shared carrier, its odd degree is

$$n_s = n_0 + 2Ars_1 + 4Ars_2 + 4|s_{\mathrm{path}}| + 2|Ds_{\mathrm{path}}|, \tag{13}$$

where D is the adjacent-difference map on the path. In particular $R \ge 6\max_j \delta_j$, and all degrees are $\Theta(r^2)$.

4.1 Positive higher-order coefficients and one dominant pair

For odd integer arguments, the hardtanh cost has the analytic extension

$$w(z) = \sqrt{\pi} z \frac{\Gamma((z+1)/2)}{\Gamma(z/2)} = \sqrt{\frac{\pi}{2}} \left(z - \frac{1}{4} + \frac{1}{32z} + O(|z|^{-2}) \right).$$

Put $c = \sqrt{\pi/2}/32$. The remainder is holomorphic and uniformly $O(|z|^{-2})$ in a sufficiently far right half-plane. Cauchy's estimate on a circle of radius $a(1-1/k)$ proves, for all $2 \le k \le \ell$ and $a \ge n_0$, once K is large enough,

$$\frac{ck!}{2a^{k+1}} \le (-1)^k w^{(k)}(a) \le \frac{3ck!}{2a^{k+1}}. \tag{14}$$

Indeed the relative remainder bound is $O(k^2/a)$, uniformly in these indices, and $n_0 \ge KA^2\ell^2/3$.

First regard the ℓ long-rail choices as independent bits t_j. If $\widetilde{W}(T)$ is the Walsh coefficient of $w(n_0 + \sum_j \delta_j t_j)$, the finite-difference formula gives

$$\frac{ck! \prod_{j\in T} \delta_j}{2^{k+1}(n_0 + D_\delta)^{k+1}} \le \widetilde{W}(T) \le \frac{3ck! \prod_{j\in T} \delta_j}{2^{k+1} n_0^{k+1}} \qquad (k = |T| \ge 2). \tag{15}$$

All singleton coefficients are negative because w is increasing. Restriction to the linear code pushes these coefficients forward:

$$W(v) = \sum_{T:\, \sum_{j\in T} a_j = v} \widetilde{W}(T). \tag{16}$$

Only zero and the individual column directions can receive a constant or singleton contribution. Every other nonzero $W(v)$ is strictly positive.

Let $u = e_1 + e_2$ in the distinguished coordinates. It is not a column, and $W(u) \ge \widetilde{W}(\{1,2\}) > 0$. The total of the other positive coefficients obeys

$$\sum_{v \ne 0, u} \max(W(v), 0) \le \eta W(u), \qquad \eta < 10^{-3}, \tag{17}$$

by increasing the fixed K if necessary. To see this quantitatively, the other independent degree-two products sum to at most $6ArS + S^2/2$, whereas the distinguished product is $8A^2r^2$. Their ratio in (15) is at most $3(6ArS + S^2/2)(1 + D_\delta/n_0)^3/(8A^2r^2) = O(A^{-1})$. For $k \ge 3$, use $k! \sum_{|T|=k} \prod_{j\in T} \delta_j \le D_\delta^k$ and sum the resulting geometric series. Relative to the distinguished pair this contributes $O(D_\delta^3/(n_0\delta_1\delta_2)) = O((KAr)^{-1})$. Pushforward cannot increase the total positive mass outside zero and u, so these bounds prove the claim.

4.2 Many distinct effective costs

For a path vector $v \in \mathbb{F}_2^r$, identify $(1,1,v) \in G$ and write

$$\nu_v = W(1,1,v) - W(0,0,v).$$

Averaging the distinguished bits gives the exact formula

$$\nu_v = -\frac{1}{2} \left[\widehat{w}_{n_0+\delta_1}(v) + \widehat{w}_{n_0+\delta_2}(v) \right], \tag{18}$$

where $\widehat{w}_a(v)$ is the path Walsh coefficient of $w(a + 4|s| + 2|Ds|)$. If $|v| \ge q \ge 5$, every set of path columns whose sum is v has at least $h = \lceil q/2 \rceil$ members. Applying (15) just to the path columns therefore proves

$$0 < -\nu_v \le \frac{3c}{n_0} \left(\frac{S}{2n_0} \right)^h. \tag{19}$$

The geometric-series factor is absorbed using $S/(2n_0) < 1/2$. Strict positivity follows because the path loop columns span the path space. This argument uses finite differences only; it does not assume an unproved positive Laplace representation for w.

Define the signed degree-count polynomial

$$P_v(X) = \sum_{s \in \mathbb{F}_2^r} (-1)^{s \cdot v} X^{2|s| + |Ds|}.$$

At least $2^{r/2}$ of these polynomials are distinct. Here is an explicit transfer-matrix certificate. For $e \in (0,1)$ set $a = 2e/(1+e^2)$ and

$$Z_v(a,e) = \sum_{y \in \mathbb{F}_2^{r-1}} e^{|y|} a^{|v+By|},$$

where B maps a set of path edges to its odd-degree vertex set. The Fourier expansion of $\exp[-t(4|s| + 2|Ds|)]$, with $e = \tanh t$ and $a = \tanh 2t$, is a common positive factor times $Z_v(a,e)$. It is also $2^{-r} P_v(e^{-2t})$. Let

$$M_0 = \begin{pmatrix} 1 & ae \\ a & e \end{pmatrix}, \quad M_1 = \begin{pmatrix} a & e \\ 1 & ae \end{pmatrix}, \quad c_0 = \begin{pmatrix} 1 \\ a \end{pmatrix}, \quad c_1 = \begin{pmatrix} a \\ 1 \end{pmatrix}.$$

Directly summing the successive edge variables gives

$$\begin{pmatrix} Z_v \\ Z_{v+e_1} \end{pmatrix} = M_{v_1} \cdots M_{v_{r-1}} c_{v_r}.$$

At $e = 1/2$, $a = 4/5$, the first matrix sends a component ratio x to $f(x) = (x + 2/5)/((4/5)x + 1/2)$, and the second sends it to $1/f(x)$. The interval $I = [4/5, 5/4]$ is invariant under both maps, and $f(I) = [20/19, 11/10] \subset (1, \infty)$, whereas its reciprocal lies below one. Both maps are injective. The resulting ratio therefore recovers v_1, then v_2, and so forth, and the terminal ratios $4/5$ and $5/4$ recover v_r. Thus all 2^r ordered pairs (Z_v, Z_{v+e_1}) are distinct. If there were fewer than $2^{r/2}$ scalar values, there would be fewer than 2^r ordered pairs. Distinct scalar values imply distinct P_v.

Distinct P_v give distinct effective costs ν_v in (18). Indeed the two degree intervals, shifted by δ_1 and δ_2, are disjoint because $\delta_2 - \delta_1 = 2Ar > S$. All degrees in both intervals lie in an interval of length $D_\delta < 2\sqrt{n_0}$ when K is large. Two distinct integers of the same squarefree part differ by more than twice the square root of the smaller. Hence these distinct degrees have distinct squarefree parts. Since $w(n)$ is a nonzero rational multiple of $\sqrt{n}$ at every odd integer n, independence of squarefree radicals forces every coefficient in a cost difference to vanish. The disjoint upper branch then forces $P_v = P_{v'}$.

4.3 An affine live set and quantitative tails

Set $q = \lceil r/64 \rceil$. For all sufficiently large r,

$$\sum_{j<q} \binom{r}{j} \le 2^{r/8}, \qquad \binom{r}{q} \le 2^{r/8}.$$

Start with one representative of each distinct polynomial, and discard representatives of weight below q. Assign to each remaining vector a q-subset of its support, and its two colors

$$\sum_{i=1}^{r} v_i \pmod 2, \qquad \sum_{i \text{ odd}} v_i \pmod 2.$$

Pigeonholing over at most $4\binom{r}{q}$ choices gives an affine fiber of size at least $2^{3r/8-3}$ in distinct-cost representatives. In particular it contains at least $2^{r/3}$ for large r. Let U be this affine fiber, with its distinguished coordinates fixed to $(1,1)$. All its path vectors have weight at least q. Put $U' = U \cup (U+u)$. Every path loop or edge column changes at least one of the two colors, and every distinguished loop changes exactly one distinguished bit. Thus

$$U' \cap (U' + a_j) = \varnothing \quad \text{for every physical column } a_j.$$

Also $0, u \notin U'$ and $u + a_j \notin U'$ for large r, since vectors in U' have at least $q > 2$ path ones. Retain one representative of each desired distinct effective cost as a live candidate, and initially reveal the other fibers in U' if necessary. The extra initial observations only impose the additional nonpositive-sign constraints already allowed in the rounding argument.

Finally (19) is $\exp[-\Omega(r \log r)]$, whereas all geometric, norm-stability, and Fourier-rounding constants are $\exp[O(r)]\,\mathrm{poly}(r)$. It therefore satisfies their strict absorption inequalities for all sufficiently large r. All parameters A, K are fixed before taking this limit.

4.4 Teacher resources

The logical target is the sign of $\sum_{i=1}^{r+2} \chi_{A_{e_i}} - (r+1)\chi_{A_0}$ and has margin one. An odd parity of degree n has an exact hardtanh spline representation using at most $n+1$ neurons: transitions centered at the successive midpoints of its attainable coordinate sums, and one constant neuron. Incoming weights have magnitude one, biases at most n, and readouts at most two. The teacher above therefore has width at most

$$C_r = (r+3)(n_0+1) + 6Ar + 8r - 4 = \Theta(r^3),$$

since the sum of path basis-vector degree increments is $8r-4$. Its readout weights are at most $2(r+1)$ and all coefficients are bounded by C_r. The physical input dimension is $d_r = 1 + 2\ell R + D_\delta = \Theta(r^2)$. The at least $2^{r/3}$ designated mistakes are consequently $\exp(\Omega(C_r^{1/3}))$.

5 Calibration, rounding, and the posterior

We now use the sharp atom theorem to identify the minimum-cost scores after each online history. This section supplies the complete argument; no choice of a favorable minimizer is substituted for a posterior claim.

In this section $\mathcal{C}$ is the retained column-code space and $\mathcal{D} = \mathcal{D}_{\mathcal{A},\mathrm{odd}}$ is its retained odd downward space from Section 3.5. Let $P_{\mathcal{C}} F = b\, g(z)$, where $g(z) = \mathbb{E}[bF \mid z]$. All projections below use the uniform physical input law. A revealed logical fiber means that all its physical inputs have been labeled. The target is positive relative to b only at $z = 0$.

Let $M = 2^m$, $m \geq 6$. Write $w_s \geq w_0 > 0$ for code weights and $W(z) = M^{-1} \sum_s w_s (-1)^{s \cdot z}$. Set $u = e_1 + e_2$. Take an affine coset $U = v_0 + H$ with $K_U = |U| \leq M/32$, where $u \notin H$ and $v_0 \notin H + \langle u \rangle$. Set $H' = H + \langle u \rangle$, $U' = U + \langle u \rangle$, and $E' = H'^{\perp} \setminus (H' + \langle v_0 \rangle)^{\perp}$. Assume $U' \cup \{u\}$ is independent in the Cayley graph whose edges add one of the code columns a_j. Section 4 constructs such a coset containing exponentially many distinct costs. Then $|E'| = M/(4K_U) \geq 8$, $s \cdot u = 0$ for $s \in E'$, and

$$\sum_{s \in E'} (-1)^{s \cdot z} = |E'| (\mathbf{1}_{H'}(z) - \mathbf{1}_{U'}(z)).$$

For every nonzero coset c of $\langle u\rangle$, put

$$k_c = \max\{0, W(z), W(z+u)\} \quad (z \in c), \qquad K_* = \sum_{c\neq 0} k_c, \qquad w_* = W(u).$$

The scalar bounds above imply, for all sufficiently large m,

$$0 < w_* \le w_0/16, \qquad K_* \le w_*/4, \qquad Mw_* \ge 16w_0, \qquad W(v+u) > W(v) \ge 0 \quad (v \in U). \tag{20}$$

5.1 The common dual

Put $L = M/2$ and define

$$h = \frac{(L-1)(w_* - 4w_0/M) + K_*}{L-2}, \qquad b_* = \frac{M(h+K_*)}{L-1}, \qquad a_* = 2w_0 - b_*.$$

These satisfy $3w_*/4 \le h \le 5w_*/4$, $b_* \le 4w_* \le w_0/4$, and $7w_0/4 \le a_* \le 2w_0$. Set

$$r_c = \frac{(L-1)k_c}{Lh + K_*}, \qquad R_* = \sum_{c\neq 0} r_c \le 1/3.$$

On $u^\perp$ define the strictly positive probability distribution

$$P(s) = \frac{1}{L}\left(1 + \sum_{c\neq 0} r_c (-1)^{s\cdot z_c}\right).$$

Its mass at zero is

$$p = P(0) = \frac{h + K_*}{Lh + K_*}, \qquad 1 - p = \frac{(L-1)h}{Lh + K_*}.$$

Let Q be P conditioned on $s \neq 0$, extended by zero outside $u^\perp \setminus \{0\}$. Define a logical dual by

$$\widehat{\lambda}(s) = w_s - 2w_0 \mathbf{1}_{s=0} - MhQ(s) - \frac{a_*}{|E'|}\mathbf{1}_{s\in E'}. \tag{21}$$

For every s, $|\widehat{\lambda}(s)| \le w_s$. Indeed its value at zero is $-w_0$, and it equals w_s if $s \cdot u = 1$. Otherwise $MhQ(s) \le Mhp/(1-p) = b_*$, so the total subtraction is at most $b_* + a_*/|E'| \le 2w_0 \le 2w_s$.

For $z \notin \{0, u\}$, Fourier orthogonality gives

$$\mathbb{E}_Q (-1)^{s\cdot z} = \frac{k_{[z]}}{h} - \frac{h + K_*}{(L-1)h}.$$

Consequently

$$\lambda(z) = M(W(z) - k_{[z]}) - a_* - a_* \mathbf{1}_{H'}(z) + a_* \mathbf{1}_{U'}(z) \quad (z \neq 0, u). \tag{22}$$

The defining equation for h also gives

$$\lambda(u) = 0, \qquad \lambda(0) = M(W(0) - w_*) \ge a_*.$$

In particular,

$$\lambda(v) = M\nu_v < 0, \qquad \lambda(v+u) = 0, \qquad \nu_v := W(v) - W(v+u) \quad (v \in U).$$

On the observed logical region $\mathcal{O} = G \setminus (U' \cup \{u\})$, its sign is positive at zero and negative elsewhere, with magnitude at least a_*.

There is also a useful strict coefficient slack. If $s \neq 0$ and $s \cdot u = 0$, then

$$\min\{w_s - \widehat{\lambda}(s), w_s + \widehat{\lambda}(s)\} \ge w_*. \tag{23}$$

For the first term use $MhQ(s) \ge 2h(1 - R_*)/(1-p) \ge w_*$. For the second use $2w_s - b_* - a_*/|E'| \ge a_*(1 - 1/|E'|) > w_*$.

5.2 Observability outside a logical independent set

Lemma 4. *If $J \subseteq G$ is independent in the Cayley graph generated by the code columns a_j, then every $A \in \mathcal{D} \cap \mathcal{C}^\perp$ satisfies*

$$\|A\|_2^2 \le 2\mathbb{E}[\mathbf{1}_{z \notin J} A^2].$$

Proof. Every Fourier support T contributing to A is an odd-degree proper subset of a carrier-containing transversal. If it had a complete rail at every pair, then either it would be a code support or, after losing the carrier, would have even degree. Thus it has an incomplete rail pair. Assign it to the first such pair i. Write $A = \sum_i A_i$ for this partition.

Conditioned on a logical fiber, two physical characters have nonzero inner product only when their symmetric difference is a union of full rail pairs with incidence vector $(a_j \cdot s)_j$ for some s. A symmetric difference containing the carrier has zero conditional expectation as well, since the carrier remains an independent uniform sign on each logical fiber. If one support is a proper subset of a rail at pair i, its complement in that pair cannot be contained in a single rail. Thus conditional covariance with another downward character requires the same restriction at pair i. It follows that A_i, A_j are conditionally orthogonal whenever $i \neq j$, and that $\ell_i(z) := \mathbb{E}[A_i^2 \mid z]$ is invariant under $z \mapsto z + a_i$. Indeed every surviving logical Fourier mode s has $a_i \cdot s = 0$.

An independent set contains at most one endpoint of each edge adding a_i. Consequently

$$\mathbb{E}[\mathbf{1}_{z \in J} \ell_i(z)] \le \|A_i\|_2^2/2.$$

Sum over i, using conditional orthogonality, to obtain the assertion. □

By the assumed Cayley independent-set property, Lemma 4 applies to $\mathcal{O}$.

5.3 Physical variance control

Suppose a score F and a nonnegative cost $B \le B_{\mathrm{cap}}$ satisfy

$$Q(F) := \sum_s w_s |\widehat{F}(A_s)| \le B, \qquad \|F\|_\infty \le H_{\mathrm{cap}}, \qquad \|F - P_{\mathcal{D}} F\|_2^2 \le S_{\mathrm{at}} B D,$$

where $D = B - \langle b\lambda, F \rangle \ge 0$ and b is the physical base character. These inequalities follow from Theorem 2. Its atomwise conditions are

$$Q(\phi_a) \le \|a\|_2, \qquad \|\phi_a - P_{\mathcal{D}} \phi_a\|_2^2 \le S_{\mathrm{at}} \|a\|_2 (\|a\|_2 - Q(\phi_a)), \quad \phi_a(x) = \mathrm{clip}(a \cdot \bar{x}),$$

for a representation $F = \sum_j c_j \phi_{a_j}$ of cost $B = \sum_j |c_j| \|a_j\|_2$. Indeed the dual inequality gives $|\langle b\lambda, \phi_a \rangle| \le Q(\phi_a)$, and Cauchy–Schwarz bounds the squared sum of residuals by $S_{\mathrm{at}} B D$. The elementary bound $|\phi_a(x)| \le \sqrt{d+1}\|a\|_2$ also gives $H_{\mathrm{cap}} = \sqrt{d+1} B_{\mathrm{cap}}$. For the retained column code, one may take $S_{\mathrm{at}} = 2^{20} \ell^2 R^2$ by the subcode estimate in Section 3.5. Assume closed full-cube margins and correct labels outside U, with logical target positive only at zero. Extra fully revealed fibers in U have the negative label. Write $P_{\mathcal{C}} F = bg$ and

$$S = \mathbb{E}[\mathbf{1}_{z \in \mathcal{O}} |\lambda(z)| (fF - 1/2)] \ge 0.$$

On each observed fiber, the variable $Y = fF$ is at least $1/2$. Thus $\mathbb{E}[|Y - \mathbb{E}Y| \mid z] \le 2(\mathbb{E}[Y \mid z] - 1/2)$. Since $|\lambda| \ge a_*$ there, this proves $\mathbb{E}[\mathbf{1}_{\mathcal{O}} |F - P_{\mathcal{C}} F|] \le 2S/a_*$. Applying independent-set observability to $P_{\mathcal{D}} F - P_{\mathcal{C}} F$ therefore yields

$$L_F := \|F - P_{\mathcal{C}} F\|_2^2 \le C_{\mathrm{phys}}(D + S), \qquad C_{\mathrm{phys}} = \max\{16 H_{\mathrm{cap}}/a_*,\ 5 S_{\mathrm{at}} B_{\mathrm{cap}}\}. \tag{24}$$

To see the constants, write $R = F - P_{\mathcal{C}} F = A + E$, where $A = P_{\mathcal{D}} F - P_{\mathcal{C}} F$ and $E = F - P_{\mathcal{D}} F$. The preceding conditional bound and $|R| \le 2H_{\mathrm{cap}}$ give $\mathbb{E}[\mathbf{1}_{\mathcal{O}} R^2] \le 4 H_{\mathrm{cap}} S/a_*$. Observability and $(R - E)^2 \le 2R^2 + 2E^2$ imply $\|A\|_2^2 \le 4\mathbb{E}[\mathbf{1}_{\mathcal{O}} R^2] + 4\|E\|_2^2$. Adding $\|E\|_2^2$ proves (24).

5.4 Linear rounding to zero or a live rectangle

Put $p_v = g(v) + 1/2$ for $v \in U$ and $\delta = D/w_* + S/a_*$. By (23), the nonconstant u-invariant part r of g satisfies $\|r\|_\infty \le D/w_*$. The region $\mathcal{O}_0 = G \setminus (U' \cup \{0, u\})$ is invariant under u, has density at least $1/2$, and consists of known negative fibers. Averaging $g + 1/2$ there eliminates the u-anti-invariant part. Consequently

$$|\widehat{g}(0) + 1/2| \le D/w_* + 2S/a_*, \qquad \sup_z |g(z) + g(z+u) + 1| \le 4\delta.$$

Each partner $v + u$ is a known negative fiber, so $p_v \ge -4\delta$. In particular the total negative mass of p is at most $M\delta$.

Compare g with

$$g_{\mathrm{ref}} = -1/2 + \mathbf{1}_0 - \mathbf{1}_u + \sum_{v \in U} p_v(\mathbf{1}_v - \mathbf{1}_{v+u}).$$

The logical ℓ^1 norm of their difference is at most

$$2MS/a_* + 4(K_U + 1)\delta \le 3M\delta.$$

Here the observed fibers contribute at most MS/a_*; the fiber u contributes at most the anchor excess plus 4δ; and the K_U partner fibers contribute at most $4K_U\delta$.

There are characters s with $s \cdot u = 1$ and $(-1)^{s \cdot v} = -1$ for every $v \in U$: prescribe the two independent linear conditions $s \cdot u = 1$ and $s \cdot v_0 = 1$ within $H^\perp$. For any such s, the dual is saturated on its positive side, so $\widehat{g}(s) \ge -D/(2w_s)$. Since

$$\widehat{g}_{\mathrm{ref}}(s) = \frac{2}{M}\left(1 - \sum_{v \in U} p_v\right),$$

we obtain

$$\sum_{v \in U} p_v \le 1 + \frac{MD}{4w_0} + \frac{3}{2}M\delta \le 1 + 2M\delta.$$

Thus the total positive mass is at most $1 + 3M\delta$.

On a fiber with $0 < p_v < 1$, the mean $g(v) = p_v - 1/2$ lies between the two allowed output intervals. A variable $Y = bF$ with $|Y| \ge 1/2$ and mean $g(v)$ has variance at least $1/4 - g(v)^2 = p_v(1 - p_v)$. Consequently

$$V_{\mathrm{frac}} := \sum_{0 < p_v < 1} p_v(1 - p_v) \le ML_F.$$

Round each p_v to its nearest $b_v \in \{0, 1\}$. Negative coordinates contribute at most $M\delta$ to the rounding distance. Coordinates above one contribute at most $3M\delta$, because total positive mass is at most $1 + 3M\delta$. The fractional coordinates contribute at most $2V_{\mathrm{frac}}$. Hence $A := \|p - b\|_1 \le 4M\delta + 2V_{\mathrm{frac}}$. If $b = 0$, choose the zero vector. Otherwise choose one coordinate with $b_v = 1$; it is live because removed fibers have $p_v \le 0$. The distance from b to this point mass is at most $3M\delta + A$. Therefore

$$\mathrm{dist}_1(p, \{0\} \cup \{e_v : v \text{ live}\}) \le 11M\delta + 4V_{\mathrm{frac}} \le B_{\mathrm{round}}(D + S), \tag{25}$$

where

$$B_{\mathrm{round}} = M(11/w_* + 11/a_* + 4C_{\mathrm{phys}}).$$

5.5 Absorption and classification of every minimum

The baseline score $g_0 = -1/2 + \mathbf{1}_0 - \mathbf{1}_u$ and rectangle scores $g_v = g_0 + \mathbf{1}_v - \mathbf{1}_{v+u}$ saturate the common dual. Their weighted Fourier costs are

$$C_0 = w_0/2 + W(0) - W(u), \qquad C_v = C_0 + \nu_v.$$

The exact calibrated decomposition is

$$B = C_0 + D + S + \sum_{v \in U} \nu_v p_v.$$

No omitted-fiber term remains: $\lambda(u) = 0$ and $\lambda(v+u) = 0$ for all $v \in U$. Let $W_{\text{eff}} = \max_{v \in U} |\nu_v|$. If $W_{\text{eff}} B_{\text{round}} < 1$, rounding gives

$$B \geq C_{\text{choice}} + (1 - W_{\text{eff}} B_{\text{round}})(D + S),$$

where the choice is zero or a live point mass. The zero choice has cost C_0, strictly greater than every live star because $\nu_v < 0$. Consequently any minimum-cost completion is a cheapest live rectangle. Keep one live representative of each distinct effective cost, revealing the remaining fibers during calibration. The live costs are pairwise distinct, so the score is unique: equality forces $D = S = 0$, then $L_F = 0$, exact anti-invariance around $-1/2$, and the rounded point mass determine every logical and physical value.

The path-code parameters in Section 4 make every constant other than $M = 2^m$ in B_{round} polynomial in m. That section constructs the affine set U with $W_{\text{eff}} \leq 2^{-3m}$ times any required fixed inverse polynomial, by imposing a sufficiently small fixed fraction of path coordinates with value one. Hence $W_{\text{eff}} B_{\text{round}} < 1$ for all sufficiently large m. The strict pairwise cost ordering proved above identifies a unique cheapest live rectangle after every history consisting of complete revealed fibers.

5.6 Every sufficiently large finite width

Lemma 5 (Integer allocation of a finite atom representation)**.** *Let $F = \sum_{j=1}^{J} c_j \operatorname{clip}(a_j \cdot \bar{x})$ be a finite representation with nonzero a_j, c_j. Put*

$$B = \sum_j |c_j| \|a_j\|_2, \qquad p_j = \frac{|c_j|}{\sqrt{d}\,\|a_j\|_2}, \qquad p = \sum_j p_j.$$

If $p < 1$, then for every sufficiently large integer N there is an exact width-N representation of F with Gaussian energy tending to $2\sqrt{d}\,B$ as $N \to \infty$.

Proof. Choose $r_j = \lceil N p_j \rceil$. For $N(1-p) \geq J$, $\sum_j r_j \leq N$. Use r_j copies of the incoming vector a_j, each with readout c_j / r_j, and pad with zero neurons. The output is exactly F. Direct calculation gives

$$K_N - 2\sqrt{d}\,B = \sum_j \frac{d\|a_j\|_2^2}{N} \frac{(r_j - N p_j)^2}{r_j} \leq \frac{d}{N} \sum_j \|a_j\|_2^2.$$

The right side tends to zero. The same statement applies after any fixed small multiplicative increase of all readouts, provided the resulting allocation sum remains less than one. □

The parity-orbit representation of bg has cost $B_w(g)$ and allocation sum

$$p = \frac{1}{\sqrt{d}} \sum_s \frac{|\widehat{g}(s)|}{\sqrt{n_s}\kappa_{n_s}}.$$

Every baseline or rectangle has ordinary Fourier norm exactly $3/2$, and $\sqrt{n}\kappa_n$ is bounded below by a positive absolute constant. Thus $p = O(d^{-1/2}) < 1/2$ for large m. Integer allocation applies to every witness and to a small strict-margin dilation of it.

Fix a history with at least one live fiber. Let C_* be its unique cheapest rectangle cost, and choose a polynomial cap B_{cap} strictly larger than all rectangle costs. The set of pairs (B, F) satisfying the closed score inequalities above, $0 \le B \le B_{\mathrm{cap}}$, $\|F\|_\infty \le \sqrt{d+1}B_{\mathrm{cap}}$, the global closed margins, and all observed labels is compact. The classification proof shows that its unique minimum-cost score is the cheapest live rectangle. At a physical input of that rectangle's exceptional fiber, intersect this compact set with the condition of a correct prediction. The resulting closed set has minimum cost $C_{\mathrm{good}} > C_*$. If it is empty, define $C_{\mathrm{good}} = +\infty$. Every representation of cost at most the cap satisfying the closed full-cube and history margins satisfies these inequalities. Representations above the cap have larger energy by (4). Therefore all feasible networks predicting correctly at the designated input have energy at least

$$E_{\mathrm{good}} = 2\sqrt{d}\,\min(C_{\mathrm{good}}, B_{\mathrm{cap}}) > 2\sqrt{d}\,C_*.$$

A small fixed dilation of the cheapest rectangle has strict margins and variation cost below $E_{\mathrm{good}}/(2\sqrt{d})$. Integer allocation realizes it at every sufficiently large width with energy below E_{good}.

Fix such a width and let $P = N(d+2)$ be the parameter dimension. A small open ball around this strict feasible wrong witness has energy at most $a < E_{\mathrm{good}}$. Choose $\rho \in (0,1)$ with $(1-\rho)E_{\mathrm{good}} > a$. The unnormalized Gaussian integral over feasible networks predicting correctly at the designated input is at most

$$e^{-(1-\rho)E_{\mathrm{good}}/(2\sigma^2)} \int_{\mathbb{R}^P} e^{-\rho K_N/(2\sigma^2)}\, d\theta = C_{N,\rho}\sigma^P e^{-(1-\rho)E_{\mathrm{good}}/(2\sigma^2)}.$$

The version-space integral is at least the ball volume times $e^{-a/(2\sigma^2)}$. Their ratio tends to zero. This proves the claimed actual posterior prediction at the designated input.

5.7 The fixed nonrepeating sequence

First reveal every physical input outside U, and every fiber in U except one chosen representative of each distinct effective cost. Sort the remaining live fibers by increasing rectangle cost. At each subsequent stage, take one fixed physical input in the cheapest surviving fiber as the designated round, then reveal the rest of that fiber before moving to the next one. All labels are target labels, and no input is repeated. The sequence depends only on the scalar costs, not on N or σ. There are at least $2^{(m-2)/3} = \exp(\Omega(m))$ designated rounds. For each fixed construction there are finitely many stages; take the maximum of their width thresholds. The posterior conclusion then holds at every stage and every fixed width beyond that maximum. Posterior-majority prediction is wrong once the correct-label probability is below $1/2$, and posterior-sampling expected mistakes converge to the designated round count by linearity of expectation.

6 Conclusion of the theorem

The path construction has $m = r + 2$ logical coordinates, $d_m = \Theta(m^2)$ physical coordinates, and a margin-one teacher of width $C_m = \Theta(m^3)$ with coefficients bounded by C_m. The affine live set contains at least $2^{r/3} = \exp(\Omega(m))$ distinct-cost representatives. The preceding posterior argument therefore gives $\exp(\Omega(C_m^{1/3}))$ designated mistakes at every sufficiently large fixed learner width. This completes Theorem 1.